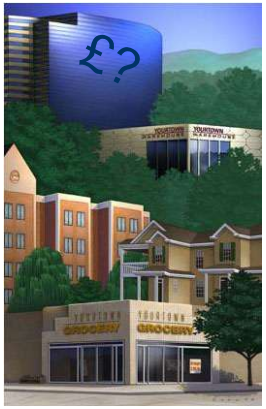
Real Estate Finance
GY462



Winter Term 2026
Prof. Olmo Silva

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GY462 — Real Estate (RE) Finance

Lecture 1

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Organizational Issues

- **Standard lectures** delivered by Prof. Olmo Silva
  - They are delivered in person and are meant to be as interactive as possible
    - Please join in the discussions and don't be shy!
    - The more we discuss, the better we learn
    - The more questions you ask, the more answers I (try to ☺) give
  - Lectures are recorded and linked on Moodle
- **Slides and other materials**: available around **a week before** the lecture
  - Try to work beforehand on the 'focus points' and be ready to discuss

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Organizational Issues – *Continued*

- **Additional lectures** (compulsory!)
  - **Week 5, Funds & Private Equity on Wednesday 6-7:30pm** – with Mr. Perry (with pre-recorded Intro by Prof. Silva)
    - (No lecture on Monday 9-11am)
  - **Week 9, Case Study I Lecture** – external interventions on RE development Monday 6-7:30pm (DFI team)
    - (No lecture on Monday 9-11am)
  - **Week 11, RE and ESG on Monday 6-7:30pm** – External lecture by Burak Kaplanoglu (RE professional)
    - (No lecture on Monday 9-11am)

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Organizational Issues – *Continued*

- **Seminars** (Dr. Ignacio Aravena Gonzalez)
  - **Weeks 2, 4, 7, 9 & 11; 4 different groups**
  - Week 11 students' presentations on **assessed** Case Study II
- **Syllabus**
  - **Lecture notes/material on GY462 Moodle page**
- **Compulsory/recommended readings**
  - **Brueggeman & Fisher (recent edition) - crucial**
  - **Linneman (recent edition) – interesting**

Organizational Issues – *Continued*

- **Financial Calculator**
  - **Used in lectures & classes (problem sets)**
  - **Strongly recommended: TI BA II PLUS**
- **LSE provides access to SNL/S&P Global**
  - **Data on thousands of properties/REITs info**
    - Useful source of data for dissertations
    - Often required to work in real estate
    - Useful to make good impression at job interviews
    - **TBC:** 'Hands-on' session organised by Dr. Pietrabissa (within GY472 – Real Estate Investment)

Assessment

- **Exam: 70% of final grade – standard format**
  - **Answer both questions (two out of two)**
  - **Similar to past exams (but only two questions)**
- **Presentation: 30% of final grade; held during classes of week 11**
  - **Work in groups to solve case-study**
  - **Provide an xl cash flow analysis as well as a presentation to be delivered during class**
  - **Assessed on both technical understanding of cash flow as well as presentation skills/pitch**

GY462 — Real Estate (RE) Finance

Introduction and Overview

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What is this Course About? *Focus 1*

- **Focus 1: Property valuation**
  - **"The value of a property is fundamentally an assessment of the associated *risks* and *opportunities*" (Linneman 2004, p. 1)**
  - **What are the downside risks and opportunities of an (income producing) property compared to 10-year Treasury note?**

Downside Risks?

1. **Operating cost**
2. **Natural disaster**
3. **Vacancy**
4. **Leasing**
5. **Liquidity**

What's the difference? Which one is a bigger risk?

Opportunities?

1. **Operating cost**
2. **Terminal value**
3. **Rental growth**

Which one is more important? Why?

Differences of Direct Real Estate to Other Financial Assets?

- **Indivisibility and large lot size**
- **Heterogeneity of assets**
- **Slow & expensive transactions**
- **Poor liquidity & private trading**

Affect risk, opportunities – or both?

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'Focus point 1'

Have you understood risks and opportunities?
Which ones stand out as more 'important'?

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What is this Course About?
Focus 2

- **Focus 2: Project financing**
- **If decide it's worth buying, how to finance the venture?**
- **Two relevant questions**
  - **(How much) should you borrow?**
  - **How?**

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What about Indirect RE investment?

- **REITS**
  - No longer part of GY462 – sorry… ☹
  - Discussed in GY472
- **MBS**
  - One lecture covering a bit of everything…
  - History of MBS, different types, structuring and pricing

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GY462 — Real Estate (RE) Finance

1. Investment Analysis

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Contents – *Chapter 1*

1. **Investment Analysis**
   - **Valuation of *Income Producing Properties***
     - **DCF Method in Real Estate Context**
     - **Choice of Discount Rate**
     - **Shortcut Methods: Cap Rate, GIM, Equivalent Yield**
   - **Pro Forma Basics**
   - **Investment Decision**
     - **Measures of Investment Performance**
     - **NPV and IRR Investment Rules**
   - **Risk Assessment Methods**

Focus of Today's Lecture

1. **Investment Analysis**
   - **Valuation of *Income Producing Properties***
     - **DCF Method in Real Estate Context**
     - Choice of Discount Rate
     - Shortcut Methods: Cap Rate, GIM, Equivalent Yield
   - Pro Forma Basics
   - Investment Decision
     - Measures of Investment Performance
     - NPV and IRR Investment Rules
   - Risk Assessment Methods

What are Income Producing Properties?

- **Properties that produce *rental* income:**
  1. **Office**
  2. **Retail**
  3. **Residential/Multifamily**
  4. **Industrial**
  5. Hotel – *why in grey?*
  6. **Others – *such as?***

Office Properties

CBD Office Space

Suburban Office Building

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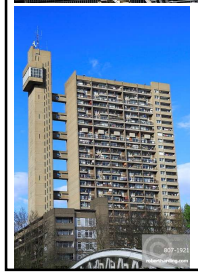
Retail Properties

Shopping centre (Westfield White City)

Shopping strip (Oxford Street)

Residential/Multifamily



High-rise apartment blocks

Small flats?

Midrise Apartments

Warehouses and Industrial Properties

Light Industrial/Office Flex Building

Warehouse interior (Amazon)

Industrial complex location

'Focus point 2'

Location, structures and tenants: what does it mean in the context of office? And in the other sectors?

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Common Demand and Supply Influences – *In Brief*

- **Demand factors: depend on the sector but crucially you should think of…**
  - **Employment, income distribution, 'agglomeration factors', local population age…**
- **Supply factors**
  - **Vacancy rates, 'available space', interest rates and financing availability, age of existing RE stock, construction/land costs…**

Discounted Cash Flow Valuation in RE Context

- **The method**
  1) **Forecast <u>expected future cash flows</u>**
  2) **Ascertain <u>required total returns</u> (discount factor or proxy for 'risk')**
  3) **Discount cash flows to <u>present value</u> at the required rate of return**

⇨ **Find value such that <u>expected return equals required return</u>** (NPV=0)

Example: Single Tenant Office Building

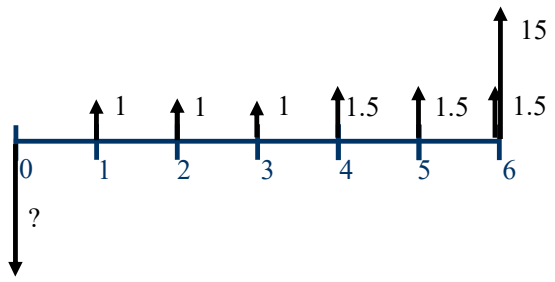
- **Assume timing:**
  - **Earn rent (6 year lease)**
  - **Trade building**
- **Expected sale price in year 6: 15**
- **Required rate of return: 10%**

Year	CF
1	1.0
2	1.0
3	1.0
4	1.5
5	1.5
6	1.5

⇨ **DCF valuation:?**

(What is the building worth today?)

DCF Diagram



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Example - *Continued*

Solution:

$FV_{t=6} = 15$

$i = 0.1$

$n = 6$

$CV_t = 1 \quad \forall\, t = 1,\ldots,3 \;\; ; \;\; CV_t = 1.5 \quad \forall\, t = 4,\ldots,6$

$$PV_{t=0} = \frac{1}{1.1} + \frac{1}{(1.1)^2} + \frac{1}{(1.1)^3} + \frac{1.5}{(1.1)^4} + \frac{1.5}{(1.1)^5} + \frac{16.5}{(1.1)^6} = 13.757$$

Example - *Continued*

Financial Calculator Solution (TI BA II Plus)

$\boxed{2nd}$ [*RESET*] $\boxed{ENTER}$ (Sets all variables to defaults)

$\boxed{2nd}$ [*FORMAT*] 3 $\boxed{ENTER}$ (3 decimal places)

$\boxed{2nd}$ [*P/Y*] 1 $\boxed{ENTER}$ (Yearly compounding) $\boxed{CE/C}$

3 $\boxed{N}$ 10 $\boxed{I/Y}$ 1 $\boxed{PMT}$ $\boxed{CPT}$ $\boxed{PV}$ PV = –2.487

3 $\boxed{N}$ 10 $\boxed{I/Y}$ 1.5 $\boxed{PMT}$ $\boxed{CPT}$ $\boxed{PV}$ PV = –3.730

3 $\boxed{N}$ 10 $\boxed{I/Y}$ 0 $\boxed{PMT}$ 3.730 $\boxed{FV}$ $\boxed{CPT}$ $\boxed{PV}$ = –2.803

6 $\boxed{N}$ 10 $\boxed{I/Y}$ 15 $\boxed{FV}$ $\boxed{CPT}$ $\boxed{PV}$ = –8.467

–2.487 $\boxed{+}$ – 2.803 $\boxed{+}$ – 8.467 $\boxed{=}$ – 13.757

'Focus point 3'

Try with your calculator:
Did you get it? Is this the only way to work it out?

What Should the Discount Rate Reflect?

- **Should reflect risk of project**
  - $r = r_f + RP$
  - r_f: **Risk free rate (Treasury bill yield; time value of money)**
  - *RP:* **Risk premium**
- **Risk premium does not reflect risk in absolute term, but contribution to risk of overall portfolio**

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What is the Correct Discount Rate?

- Specific to type of property and its contribution to portfolio
- Depends on risk characteristics of project
- Reflects cost of foregoing alternative investments of similar nature
  - Opportunity cost of capital

Sources of Discount Rate?

Discount Rate and Risk Premium for Longer Term Projections

- Typical 'real world' commercial property analysis uses <u>10 year projection of CFs</u>
- Can we assume fixed discount rate like in previous example?
  - What if current lease expires in year 6?
  - Initial lease has lower risk
    - Existing contract → known cash flows
    - Risk limited to tenant default risk
  - Economic situation when lease is expiring?

Example with Lease Renewal

- Consider 10 year projection
- First 6 years: existing lease
- Next 4 years: expect new lease (with uncertainty)
- Expected sale price in year 10: 20
- Required rate of return:
  - 8% first 6 years
  - 15% next four years (more risky)

⇨ DCF valuation in year 2000?

Year	CF
2001	1.0
2002	1.0
2003	1.0
2004	1.5
2005	1.5
2006	1.5
2007	2
2008	2
2009	2
2010	2

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Example with Lease Renewal – *Continued*

Solution:

$FV_{t=10} = 20$

$i_t = 0.08 \;\; \forall \;\; t=1,...,6 \;\; ; \;\; i_t = 0.15 \;\; \forall \;\; t=7,...,10$

$n = 10$

$CV_t = 1 \;\; \forall \; t = 1,...,3 \;\; ; \;\; CV_t = 1.5 \;\; \forall \; t = 4,...,6 \;\; ;$

$CV_t = 2 \;\; \forall \; t = 7,...,10$

$$PV_{t=0} = \sum_{t=1}^{3} \frac{1}{(1.08)^t} + \sum_{t=4}^{6} \frac{1.5}{(1.08)^t} + \sum_{t=7}^{10} \frac{2}{(1.15)^t} + \frac{20}{(1.15)^{10}}$$

$$= 13.058$$

Example with Lease Renewal – *Continued*

⇨ **Is this really the 'only' solution?**

Intra/Inter-Lease Discount Rate

- **Distinguish between CFs covered by signed lease and CFs/sales price expected but not guaranteed**
  - ***Intra-lease* (within) discount rate: Reflects tenant default risk**
  - ***Inter-lease* (between) discount rate: Reflects rental market risk/uncertainty about sales price**

⇨ **What is the correct DCF-value in this case?**

Solution – *Step by Step*

- **Risk components?**
  - **Risk associated with initial lease**
    - **Limited to tenant default risk**
    - **CFs are known (existing contract)**
  - **Risk associated with second lease**
    - **Economic situation when lease is expiring?**
    - **Once 2nd lease is signed: tenant default risk?**

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Solution

Step 1: Expected PV in t=2000 of first lease (6 yrs CF)

$CV_t = 1 \ \forall \ t = 2001,\ldots,2003 \ ; \ CV_t = 1.5 \ \forall \ t = 2004,\ldots,2006$

$$PV_{t=2000} = \sum_{t=2001}^{2003} \frac{1}{(1.08)^{t-2000}} + \sum_{t=2004}^{2006} \frac{1.5}{(1.08)^{t-2000}}$$

$= 2.577 + 3.069 = 5.646$

Intra-lease discount rate

Solution – *Continued*

Step 2: Expected PV in t=2006 of second lease (4 yrs CF)

(Consider 2nd lease as if it were 2006 and lease with expected conditions is signed...)

$CV_t = 2 \ \forall \ t = 2007,\ldots,2010$

$$PV_{t=2006} = \sum_{t=2007}^{2010} \frac{2}{(1.08)^{t-2006}}$$

$= \underline{6.624}$ (value in 2006 not 2000!)

Intra-lease discount rate

How much is 6.624 in 2000?

Solution – *Continued*

Step 3: Expected PV of second lease in t=2000

FV of second lease: 6.624 (in t=2006)

PV of second lease in t=2000: ?

Discount rate? $\Rightarrow$ 15% (Value more uncertain)

$$PV_{t=2000} = \frac{6.624}{(1.15)^{2006-2000}}$$

$= \underline{2.864}$ (value in 2000!)

Inter-lease discount rate

Solution – *Continued*

Step 4: Expected PV of property sale in 2010?

FV of property: 20 (in t=2010)

PV of property in t=2000: ?

Discount rate? $\Rightarrow$ 15% (Interlease discount rate!)

$$PV_{t=2000} = \frac{20}{(1.15)^{2010-2000}}$$

$= \underline{4.944}$ (value in 2000!)

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Solution – *Continued*

Step 5: DCF value of property in 2000?

PV of 1st lease in t=2000: 5.646
PV of 2nd lease in t=2000: 2.864
PV of sale in year 10 in t=2000: 4.944

DCF Valuation in 2000: 5.646+2.864+4.944=13.454

Intra/Inter-Lease Discount Rate – *Summary*

⇨ **Alternative valuation to better reflect risk:**

Intra-lease discount rate: risk *during lease-term* reflecting risk that tenant might default on contractual lease obligation

13.453 =

$$PV_{t=0} = \sum_{t=1}^{3} \frac{1}{(1.08)^t} + \sum_{t=4}^{6} \frac{1.5}{(1.08)^t} + \left(\frac{1}{(1.15)^6}\right) \sum_{t=1}^{4} \frac{2}{(1.08)^t} + \frac{20}{(1.15)^{10}}$$

Inter-lease discount rate: higher risk reflecting uncertainty about future lease contract (rental market risk) and uncertainty about sales price

'Focus point 4'

**Intra- vs. inter- discount factor;
tenancy vs. lease risk:
Are these terms clear? How are they related?**

Conclusion?

⇨ **Given similar estimates of**

- CFs and sales price at end of projection
- Tenant default risk and discount rate associated with uncertain future economic conditions

▪ **Sophisticated investor can earn a fortune**

- In example: £396,000!
- This is the difference between simple price (13.058) and the inter-intra price (13.454)

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In Practice – *Some caveats*

- Complicating factors
  - Variety of leases (and lease-expiry dates) within same property
  - Cash flows not entirely fixed: uncertainty on operating expenses and other determinants of cash flow

➔ Mechanical, but sometimes complex, 'matching' exercise!

In Practice – *Continued*

- Moreover, uncertainty over correct discount rates
  - Explicit intra-lease and inter-lease discount rates not common
  - Industry often takes overall view of risk: the 'blended' discount rate or internal rate of return (IRR)
    - More on IRR next lecture and classes

Alternative valuation methods?

- Appraisal based on income produced by property not unique
- *Sales comparison* approach
  - Analyze data of recently sold highly comparable properties
  - Adjust for differences
  - Rationale? No-one would pay more for a property than other recent investors

Alternative valuation methods? *Continued*

- *Construction cost* approach
  - Analyze construction costs and cost of buying land
  - For existing buildings: discount physical, functional and external obsolescence
  - Rationale? No-one would pay more for a property than it would cost to build it
- Various methods should give similar values if well implemented

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Outlook: Next Lecture

- Continue with valuation methods based on income
- Present short cut valuation methods
- Look at Pro Forma
  - Multi-year CF forecast
  - Operating CF & Reversion CF
- Conclude with RE risk assessment methods

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